UNCLASSIFIED // OPEN SOURCE ACADEMIC RESEARCH

IRAN WEAPONS OF MASS DESTRUCTION

COMPREHENSIVE THREAT ASSESSMENT

Nuclear · Biological · Chemical · Delivery Systems

Pre-War, Wartime & Post-Strike Status | Intelligence Collection Requirements | Scientific Personnel Dossiers

Assessment Date: March 2026

Source Basis: Open-Source Intelligence, IAEA Reports, [TIER 1] Congressional Research Service, ISIS (Institute for Science and International Security), CSIS, [TIER 2] Arms Control Association, NTI, Iran Watch, UANI, Academic Literature

NOTICE	*This document is compiled entirely from open-source, publicly available academic and government sources. It contains no classified information. It is intended for academic nonproliferation research only. All coordinates, personnel data, and facility information are drawn from published IAEA reports, congressional testimony, peer-reviewed literature, and established nonproliferation research organizations.*

— EXECUTIVE SUMMARY —

ANALYTICAL METHODOLOGY AND CONFIDENCE FRAMEWORK: This assessment is an open-source analytic synthesis drawn from publicly available government reports, IAEA documentation, peer-reviewed academic literature, established nonproliferation research organizations, and media reporting of varying quality. Confidence levels vary by issue set and are noted at point of use. Denied IAEA access to Iran since June 2025 materially limits independent verification. Key judgments are inferential and scenario-based where noted; not all conclusions rest on direct evidentiary confirmation. Three source tiers are used: Tier 1 = U.S. Government assessments, IAEA reports, Congressional Research Service, peer-reviewed literature. Tier 2 = Established nonproliferation research organizations (ISIS, CSIS, Arms Control Association, NTI, Iran Watch, UANI). Tier 3 = Media reporting, opposition-sourced intelligence (NCRI), single-source or anonymous-source claims. Tier 3 sources are identified as such at point of use. No Tier 3-sourced claim is presented with Tier 1 certainty.

EXECUTIVE SUMMARY

This summary organizes findings into four categories used throughout the document: OBSERVED (reported facts from named sources), ASSESSMENT (analytic inference from available evidence), KEY UNKNOWN (unresolved questions), and IMPLICATION (why the gap matters strategically).

OBSERVED: The June 2025 U.S.-Israeli strikes (Operation Midnight Hammer) destroyed significant declared enrichment infrastructure at Natanz, Fordow, and Isfahan. As of March 2026, IAEA inspectors have not been permitted to re-enter Iranian nuclear sites. The IAEA confirmed, prior to losing access, that Iran held approximately 408–440 kilograms of uranium enriched to 60% U-235 — the largest such stockpile held by any non-nuclear-weapon state. The 2025 ODNI Annual Threat Assessment (Tier 1) elevated Iran's assessed offensive biological weapons program from 'probably' to 'very likely.' Commercial satellite imagery indicates continued construction and hardening activity at Pickaxe Mountain.

ASSESSMENT: The June 2025 strikes degraded Iran's declared enrichment capacity but did not eliminate the fissile material required for multiple nuclear devices. Available reporting is consistent with post-strike dispersal and reconstitution activity. Iran's scientific and engineering knowledge base is assessed as substantially intact and distributed across multiple institutions. The operational distinction between 'not building a nuclear weapon' and 'possessing prerequisites for rapid production' appears to have narrowed; the Intelligence Community assessed through mid-2025 that no formal decision to build a weapon had been made. The degree to which this assessment remains current post-June 2025 is an open question.

THREE CRITICAL UNKNOWNS: (1) The physical location and integrity of approximately 440 kg of 60%-enriched uranium — unverified by any international body since June 2025. (2) The interior configuration and operational status of Pickaxe Mountain — IAEA access has never been granted to this site. (3) Whether the Kavir Plan (NCRI, June 2025 — Tier 3 source, unconfirmed) represents an active, organized weaponization program in Semnan Province.

IMPLICATION: Every strategic option — military, diplomatic, or verification-based — is materially impaired without resolution of Unknown (1). Breakout timelines in Section 1.4 are conditional on stockpile accessibility; if the material is inaccessible or dispersed, those timelines lengthen. If it is accessible and a covert cascade is operational, they compress significantly. The range of plausible outcomes is wide precisely because the most consequential variable is currently unobserved.

KEY JUDGMENT	*Under assessed worst-case conditions — covert centrifuge cascade operational, 60% stockpile accessible — available technical analysis indicates Iran could produce enough weapons-grade uranium for one nuclear device in approximately 5–10 days (Tier 1: DIA, May 2025). These conditions are assessed as possible but not confirmed. Assembly of a crude deliverable device would require an additional 2–4 months. A missile-deliverable warhead would require 1–2 years from decision. The primary intelligence gap is the location of the 440 kg stockpile and the operational status of undeclared centrifuge*

capacity.

PART I: NUCLEAR WEAPONS PROGRAM

1.1 Program Overview and Current Status

Iran's nuclear weapons program traces to the late 1980s when the Islamic Revolutionary Guard Corps initiated efforts to develop nuclear capability following the Iran-Iraq War. The formal weapons development effort, known as the Amad Plan, ran from approximately 1989 through 2003. It was formally halted in 2003 following exposure of undeclared facilities at Natanz and Arak. The U.S. Intelligence Community has consistently assessed the weapons program was halted in 2003 and has not formally resumed as an organized, structured program. However, the 2025 ODNI Annual Threat Assessment notably omitted language from prior years stating Iran was 'not building a nuclear weapon,' and Iran was assessed to be conducting activities that 'better position it to produce' nuclear weapons.

The operational distinction between 'not building a nuclear weapon' and 'possessing all prerequisites for rapid production' has narrowed significantly. The Intelligence Community continued to assess through mid-2025 that no formal decision to build a weapon had been made; the degree to which this remains the case post-June 2025 is an active analytical question. As of mid-2025, Iran had the fissile material for nine or more nuclear weapons if further enriched, operating centrifuge cascades capable of completing that enrichment in days, and a scientific establishment with decades of weapons-relevant design experience.

1.2 Site Inventory: Coordinates, Status, and Intelligence Gaps

FACILITY	COORDINATES	STATUS (MAR 2026)	IAEA ACCESS
Natanz FEP	33.7252°N 51.7272°E	**DAMAGED — INACCESSIBLE**	*Blocked since June 2025*
Fordow FFEP	34.8826°N 50.9999°E	**DAMAGED — STATUS UNCLEAR**	*Blocked since June 2025*
Isfahan UCF + IFEP	32.5807°N 51.8277°E	**HEAVILY DAMAGED**	*IAEA: location of IFEP unknown*
Pickaxe Mountain	~33.707°N 51.718°E	**ACTIVE CONSTRUCTION**	*Never granted*
Parchin: Taleghan 1&2	35.5275°N 51.7523°E	**ACTIVE — BEING HARDENED**	*One brief visit; incomplete*
Parchin: Golab Dareh	35.4955°N 51.7442°E	**STATUS UNKNOWN**	*Never visited*
Parchin: Shahid Boroujerdi	~35.515°N 51.760°E	**STATUS UNKNOWN**	*Never visited*
Arak IR-40 Reactor	34.3704°N 49.2411°E	**OPERATIONAL (modified)**	*Under safeguards*
Turquzabad (Loc. 1)	~35.618°N 51.803°E	**INVESTIGATION OPEN**	*Limited — uranium found*
Varamin (Location 3)	~35.314°N 51.668°E	**INVESTIGATION OPEN**	*Limited — uranium found*
Marivan Test Site	~35.521°N 46.179°E	**STANDBY / PERIODIC**	*One visit; inadequate*
Lavizan-Shian (Razed)	35.773°N 51.498°E	**DEMOLISHED 2003-2004**	*Post-demolition only*
Lashkar Ab'ad Laser	35.874°N 50.764°E	**INACTIVE**	*Inspected 2014*

Kalaye Electric Co.	~35.733°N 51.567°E	**HISTORICAL — INACTIVE**	*Post-sanitization only*
Chalus (Suspected)	~36.650°N 51.417°E	**UNKNOWN**	*Never visited*
Kolahdouz (Suspected)	~35.733°N 50.850°E	**UNKNOWN**	*Never visited*
Saghand Uranium Mine	32.550°N 55.230°E	**OPERATIONAL**	*Declared — safeguarded*
Ardakan Processing Plant	32.308°N 54.015°E	**OPERATIONAL**	*Declared — safeguarded*
Bushehr NPP	28.830°N 50.886°E	**OPERATIONAL**	*Russian-managed safeguards*
Kavir Plan Sites (est. 6)	Semnan Province (undisclosed)	**ALLEGED — UNCONFIRMED**	*None — undeclared*

NOTE ON PICKAXE MOUNTAIN	*Kuh-e Kolang Gaz La (Pickaxe Mountain) represents the highest-priority unresolved site. Construction has accelerated since the June 2025 strikes. Tunnel entrances are being extended and covered with concrete and gravel. Assessed depth exceeds Fordow (est. 260–330 feet). Iran claims centrifuge assembly. Some analysts assess that it may hold diverted enriched uranium stockpiles or operational centrifuge cascades. This inference is circumstantial; the interior configuration is unknown. IAEA has never been granted access.*

1.3 The 60% Stockpile — Location Assessment

The central unresolved question in Iran's nuclear program is the physical location and integrity of approximately 440 kg of uranium enriched to 60% U-235 (uranium mass basis), the largest disclosed stockpile of near-weapons-grade uranium outside declared nuclear weapons states. This material represents Iran's primary strategic nuclear asset.

Assessed Location Scenarios (Confidence Rankings):

- SCENARIO A — Remains on-site at damaged facilities [MODERATE CONFIDENCE]: IAEA's November 2025 assessment held that material had not departed Esfahan, Fordow, or Natanz sites, but acknowledged it may have moved within sites. UF6 cylinders require specialized handling; rapid unsupported evacuation carries contamination risk. Assessment weakened by lack of inspector access.
- SCENARIO B — Relocated to Pickaxe Mountain [MODERATE-HIGH CONCERN]: Acceleration of construction post-strikes is consistent with active use. Location 1 mile south of Natanz enables covert transfer via underground connecting infrastructure. Depth provides protection against future strikes. Material could have been moved pre-strike during intelligence warnings.
- SCENARIO C — Dispersed to IRGC underground military sites [POSSIBLE]: Iran's IRGC maintains extensive hardened underground facilities separate from declared nuclear sites. Small quantities dispersed across multiple sites would survive any single strike and evade detection. Consistent with Iran's demonstrated dispersal doctrine.
- SCENARIO D — Active enrichment to 90% at undeclared site [ASSESSED POSSIBLE]: The Kavir Plan (NCRI, June 2025) alleges six sites in Semnan Province. Even a small cascade of 9 IR-6 centrifuge cascades fed by the 60% stockpile could produce one weapon's worth of HEU in 5–10 days. The NCRI has a credible track record (correctly identified Natanz 2002, Fordow 2009) but also has produced unreliable intelligence.

1.4 Breakout Timeline Analysis

The following timeline represents the scientific and engineering sequence required from stockpile access to a functional nuclear device. Figures are drawn from the [TIER 1] Defense Intelligence Agency (May 2025), Arms Control Association, ISIS, Congressional Research Service, and published enrichment physics calculations.

PHASE	PROCESS	TIMELINE (STOCKPILE INTACT)	TIMELINE (POST-STRIKE)
60% → 90% HEU (1 bomb)	Centrifuge re-enrichment of UF6	**5–10 DAYS**	**6–10 WEEKS**
UF6 gas → uranium metal	Chemical reduction, casting	2–4 weeks	2–6 weeks
Crude gun-type device	Simpler; no implosion needed; truck/ship deliverable	2–6 weeks	2–3 months
Implosion device (untested)	Precision explosive lenses; hydrodynamic testing needed	3–6 months	6–12 months
Missile-deliverable warhead	Miniaturization; re-entry vehicle; Shahab-3/Sejjil integration	1–2 years	2+ years

Note on gun-type pathway: U.S. intelligence assessed in late 2024 that Iran was exploring a cruder gun-type fission weapon, undeliverable by missile, which could be manufactured in a few weeks. This represents a meaningful doctrinal shift — prioritizing rapid deterrence-by-possession over a sophisticated deliverable warhead. A gun-type device requires approximately 50 kg of HEU (vs. 16–25 kg for implosion), meaning the existing 440 kg stockpile contains sufficient material for roughly 8 gun-type devices if enriched to 90%.

1.5 Key Nuclear Scientific Personnel

Iran's nuclear weapons program has been substantially indigenized since the 1990s. The following figures represent publicly identified key personnel based on IAEA documentation, UN Security Council designations, U.S. Treasury sanctions listings, and academic records.

SCIENTIST	ROLE	EDUCATION	INSTITUTIONAL AFFILIATION	STATUS
Mohsen Fakhrizadeh	Head, AMAD Plan; SPND Director; Weapons Program Lead	BSc Nuclear Physics: Shahid Beheshti Univ. PhD (Nuclear Radiation/Cosmic Rays): University of Isfahan	IRGC; Imam Hossein University (professor); Physics Research Center (Lavizan)	**ASSASSINATED Nov. 27, 2020 (Mossad attributed)**
Ali Akbar Salehi	Head, Atomic Energy Organization of Iran (AEOI); key JCPOA negotiator	PhD Nuclear Engineering: MIT (Massachusetts Institute of Technology), USA	AEOI; Sharif University of Technology; UN Ambassador	**ACTIVE — Senior advisor role post-AEOI**
Fereydoon Abbasi-Davani	Former AEOI head; weapons-	PhD Nuclear Physics: Shahid Beheshti	AEOI; Imam Hossein University;	**KILLED — June 13, 2025 Israeli**

	relevant physics research; survived 2010 assassination attempt	University, Tehran	IRGC-affiliated research	**strikes**
Ardeshir Hosseinpour	Electromagnetic separation; uranium isotope work	Shiraz University; Malek Ashtar University of Technology, Isfahan	Shiraz University; Malek Ashtar University	**DIED Jan. 2007 — suspected radiation poisoning or assassination**
Ahmadreza Zolfaghari	Nuclear physics professor; Faculty dean	Faculty of Nuclear Sciences: Shahid Beheshti University, Tehran	Shahid Beheshti University (dean, nuclear faculty)	**KILLED — June 2025 Israeli strikes**

1.6 Educational Infrastructure and Self-Sufficiency Assessment

Iran's nuclear program is now substantially self-sustaining at the scientific level. The key institutions that train Iran's nuclear workforce include:

- Shahid Beheshti University (Tehran) — Primary undergraduate and graduate nuclear physics training; faculty of nuclear sciences. Named after a revolutionary martyr; deeply integrated with IRGC academic networks.
- University of Isfahan / Isfahan University of Technology — Graduate nuclear physics and engineering. Fakhrizadeh received his PhD here. Strong centrifuge and materials physics programs.
- Imam Hossein University (Tehran) — IRGC-operated military university. Provides cover for military-affiliated nuclear research personnel. Fakhrizadeh was a physics professor here.
- Sharif University of Technology (Tehran) — Iran's premier technical institution. Salehi trained at MIT but Sharif produces the domestic nuclear engineering pipeline. Equivalent in status to MIT for Iranian domestic engineering.
- Malek Ashtar University of Technology (Isfahan/Tehran) — Defense-affiliated technical university. Provides dual-use research environment for nuclear and conventional weapons engineering.
- Amirkabir University of Technology (Tehran) — Nuclear and materials engineering programs with defense contractor ties.

The Department of Defense early assessments noted Iran's dependence on foreign expertise and encouraging students to study abroad. The degree of remaining foreign technical dependence is assessed as substantially reduced, though not fully eliminated. Iran now has multiple generations of domestically trained nuclear scientists. The assassination campaign (2010–2025) killed or wounded over a dozen scientists but, per AEOI head Salehi's own statement, prompted more students to enter nuclear fields as a patriotic response. Iran's knowledge base is now distributed across hundreds of scientists and engineers, not dependent on any single individual or institution.

Foreign Technical Acquisition: In the 1990s, Iran obtained centrifuge designs and components through the A.Q. Khan proliferation network (Pakistan). Khan provided P-1 and P-2 centrifuge technical drawings, components, and critical tolerances. This transferred a decade of Pakistani centrifuge development experience directly to Iran, substantially accelerating its program. Russia provided the Bushehr reactor and associated training of Iranian nuclear engineers, building domestic expertise in reactor physics and fuel cycle management. China provided uranium conversion technology for the Isfahan UCF (a Chinese design), and Chinese dual-use equipment suppliers were sanctioned by the U.S. in 2002 for providing materials to Iran's biological and chemical programs.

PART II: BIOLOGICAL WEAPONS PROGRAM

2.1 Assessment and Confidence Level

2025 ODNI ASSESSMENT	*The 2025 U.S. Intelligence Community Annual Threat Assessment elevated Iran's offensive biological weapons program probability from 'probably' to 'very likely.' This represents a meaningful upward revision. The ODNI characterization suggests accumulated intelligence pointing toward active offensive development, though the specific intelligence basis is not disclosed in the unclassified assessment.*

The U.S. Department of Defense has assessed Iran as a biological weapons concern since at least the mid-1990s. The Arms Control and Disarmament Agency assessed in 1996 that Iran had probably produced biological warfare agents and apparently weaponized a small quantity. Iran is a signatory to the Biological and Toxin Weapons Convention (BTWC) but has not been in compliance with its obligations.

Iran International reported in January 2026 (Tier 3 — anonymous military sources, single outlet) that the IRGC Aerospace Force may be developing biological and chemical warhead integration for ballistic missiles, including possible optimization of delivery systems and associated command infrastructure. If accurate, this would represent a qualitative escalation from agent production toward warhead integration — though this report rests on anonymous sources and has not been independently verified.

2.2 Known and Suspected Biological Agent Capabilities

AGENT	ASSESSED STATUS	DELIVERY ASSESSED	SOURCE / BASIS
Anthrax (B. anthracis)	**VERY LIKELY developed; Material Threat Determination issued by U.S. gov.**	Aerosol; ballistic missile warhead integration underway	ABAC Fact Sheet 2025; ODNI ATA 2025; DoD assessments 1996–2012
Smallpox (Variola)	**POSSIBLE — former Soviet Biopreparat stockpiles potentially accessible via Russian state actors or black market**	Aerosol; requires sophisticated stabilization	Brookings BWC analysis; Alibek defector testimony on Soviet legacy stocks
Botulinum toxin	PROBABLE — dual-use pharmaceutical production infrastructure provides cover	Aerosol; food contamination; missile warhead	DoD Proliferation: Threat and Response assessments; ACDA 1996
Plague (Y. pestis)	POSSIBLE — assessed as within technical reach; endemic in regional rodent populations	Aerosol; flea vector; warhead	CRS; GlobalSecurity assessments

2.3 Foreign Assistance to Iran's Biological Program

Russia / Soviet Legacy (Biopreparat)

The Soviet Union operated the world's largest offensive biological weapons program (Biopreparat), employing 60,000+ scientists at its peak. After the Soviet collapse, hundreds of Biopreparat scientists faced unemployment. Iran recruited from this pool in the 1990s. Ken Alibek, former Biopreparat deputy director and key defector, provided testimony to U.S. authorities about the scale of post-Soviet proliferation risk. Open-source reporting and defector testimony indicate Iran likely benefited from Soviet-era expertise relevant to aerosol weaponization and agent stabilization, though the precise scope and current residual effect of this assistance is not independently verifiable.

China

In the 1990s, U.S. Secretary of State Madeleine Albright raised specific concerns about transfers of dual-use items from Chinese entities to Iran's biological weapons program. The U.S. sanctioned three Chinese firms in January 2002 for supplying materials used in manufacture of chemical and biological weapons. The State Department's 2021 report assessed China likely maintained an offensive BW program after signing the BTWC in 1984, and expressed concern about continued transfers of controlled items to nations including Iran. Chinese dual-use fermentation, growth media, and containment equipment exports to Iran have been cited in multiple proliferation assessments.

North Korea

North Korea and Iran have maintained substantial missile cooperation since the 1990s (Scud-B/C transfers, Nodong/Shahab-3 development). BW-specific technical cooperation is less well documented in open sources, but the two states share doctrine around asymmetric deterrence and have established procurement channels that could facilitate dual-use biological equipment transfers. North Korea is assessed by the U.S. as maintaining an active offensive BW program.

2.4 Suspected Biological Weapons Facilities

FACILITY / LOCATION	COORDINATES (APPROX)	ASSESSED FUNCTION	BASIS FOR ASSESSMENT
Damghan Facility	~36.10°N 54.20°E	Biological/chemical weapons-related R&D; also listed as chemical warhead production site in declassified US databases	Johnston's Archive declassified US WMD facility database; US government assessments
Imam Khomeini Research Center, Karaj	~35.79°N 51.00°E	NCRI alleged BW research including anthrax agent work; proximity to nuclear research center at Karaj; dual-use pharmaceutical and agricultural research declared	NCRI press conference 2003; Iran Watch reporting
Marzban Biotech Complex	Tehran area (undisclosed)	NCRI alleged BW agent production under pharmaceutical cover; not independently confirmed	NCRI; Iran Watch; LOW CONFIDENCE — NCRI source
IRGC Aerospace BW Warhead Program (Eastern Iran)	Eastern Iran (specific locations undisclosed; IRGC missile bases)	Reported (Iran International, anonymous military sources, January 2026) as possible BW and CW warhead integration work. Missile launcher transfers to eastern regions noted concurrent with these reports. Described by sources as 'complementary deterrent factor'	Iran International (anonymous military sources), Jan. 2026; corroborated by ODNI 2025 BW assessment elevation

PART III: INTELLIGENCE COLLECTION REQUIREMENTS AND ANALYTICAL GAPS

3.1 Priority Intelligence Requirements (PIRs)

The following represent the most critical unanswered questions driving collection requirements, ranked by strategic urgency based on open-source assessments from the Congressional Research Service, ISIS, and CSIS:

PRIORITY	INTELLIGENCE REQUIREMENT	COLLECTION METHOD	CURRENT GAP
P1	**Physical location and container integrity of 440 kg of 60% enriched uranium stockpile**	HUMINT; SIGINT; SAR satellite ground deformation; antineutrino detection (emerging)	**CRITICAL — unresolved since June 2025**
P2	**Operational status of Pickaxe Mountain — does it contain centrifuge cascades and/or fissile material?**	Commercial SAR (InSAR ground deformation); thermal IR imaging; atmospheric UF6 sampling; HUMINT	**CRITICAL — IAEA never granted access**
P3	**Number, location, and type of undeclared centrifuges built but not installed pre-2025 and their current storage locations**	SIGINT (procurement/logistics networks); HUMINT (manufacturing workers); satellite imagery of warehousing	HIGH GAP — IAEA lost monitoring in 2021
P4	**Status and scope of Kavir Plan (Semnan Province) — is there active weaponization work at six undeclared sites?**	HUMINT (NCRI network; internal IRGC sources); satellite imagery; SIGINT; hydrodynamic test detection (seismic)	UNCONFIRMED — NCRI alleged only
P5	**Status of Taleghan-2 reconstruction at Parchin — is active weaponization physics work resuming?**	Commercial satellite imagery (CSIS, ISIS); seismic monitoring for hydrodynamic tests; HUMINT	PARTIAL — imagery shows hardening; interior unknown
P6	**Identity, location, and protection status of surviving nuclear weaponization scientists post-June 2025 strikes and counterintelligence crackdown**	SIGINT; biometric/facial recognition satellite; HUMINT; academic publication monitoring	HIGH GAP — Iran executed 700+ suspects post-strikes; HUMINT penetration reduced

3.2 Collection Methods and Technical Applicability

Satellite Imagery (Electro-optical and SAR)

Commercial providers (Maxar, Planet, Airbus) provide sub-meter resolution imagery used by CSIS, ISIS, and Washington Post for Pickaxe Mountain and Natanz damage assessments. InSAR (Interferometric Synthetic Aperture Radar) can detect ground deformation from underground excavation and construction — directly applicable to monitoring tunnel development at Pickaxe Mountain and the 2007 Natanz tunnel

complex. Thermal infrared can detect heat signatures from centrifuge friction and motor operation through ventilation exhaust. Iran has progressively hardened facilities against this by going deeper underground.

Environmental Sampling and Atmospheric Detection

Downwind air sampling for uranium hexafluoride and its decay products can provide evidence of active enrichment at undeclared sites. The IAEA's clean laboratory in Seibersdorf can detect individual uranium particles at concentration levels near zero from swipe samples. The limitation is access — this requires either IAEA inspection rights or assets positioned near suspected sites. For Pickaxe Mountain specifically, mobile air sampling stations in the Natanz area could provide detection if processing activities are generating atmospheric signatures. The detection probability is higher for active enrichment than for stored UF6 cylinders.

Seismic Monitoring

Hydrodynamic testing at Parchin (Taleghan-2, Marivan) generates detectable seismic signatures distinguishable from earthquakes using P-wave/S-wave ratio analysis. The CTBTO's network of 170 seismic stations plus regional networks can detect test events above roughly 10-50 tons of conventional explosive equivalent. This is directly relevant to monitoring implosion device development at Parchin. Available technical analysis indicates that weapons-grade hydrodynamic testing would be expected to generate detectable seismic signatures, though detection thresholds vary by test geometry and depth.

Antineutrino Detection (Emerging Capability)

DARPA's QuSeN program is developing lightweight, mobile antineutrino detectors. Nuclear fission produces antineutrinos in proportion to reactor power that cannot be shielded. A covert centrifuge cascade does not produce significant antineutrino flux — but any active nuclear reactor or criticality experiment would. This technology, once deployable, would allow standoff monitoring of covert reactor operation (relevant to any plutonium pathway) but does not directly address centrifuge-based HEU production.

SIGINT and Procurement Network Monitoring

Iran's centrifuge program depends on controlled dual-use materials: maraging steel (for rotor tubes), carbon fiber composites, specialty bearings, frequency converters, vacuum pumps. Monitoring procurement networks — front companies, third-country intermediaries, shipping manifests — provides advance warning of reconstitution activity. Iran's procurement networks have been substantially disrupted by sanctions but historically re-emerge through new intermediaries in Turkey, UAE, and East Asian states.

3.3 The Confidence Assessment Framework

Intelligence confidence levels in nonproliferation assessments are assigned based on the following scientific and methodological criteria, as described in IAEA and ODNI reporting standards:

- HIGH CONFIDENCE: Multiple independent collection methods producing consistent results. Physical evidence (environmental samples, satellite-confirmed construction) combined with documentary evidence (nuclear archive captures) and corroborating HUMINT. Example: Fordow enrichment violation — confirmed by IAEA environmental samples, centrifuge installation count, facility design documentation, and isotope detection.
- MODERATE CONFIDENCE: Two or more collection methods, but with potential ambiguity in interpretation. Satellite imagery showing activity consistent with but not exclusive to assessed function. Example: Pickaxe Mountain as centrifuge assembly — consistent with all observed indicators but interior not confirmed.
- LOW CONFIDENCE (POSSIBLE): Single-source or unconfirmed intelligence. NCRI allegations fall here unless independently confirmed. Kavir Plan (Semnan) currently assessed at low confidence. The NCRI has a mixed but non-trivial track record.
- UNKNOWN / UNRESOLVED: Specific questions where available collection methods cannot provide an answer without access. Location of 440 kg stockpile falls in this category. No remote

sensing method can confirm UF6 cylinder location in hardened underground facility without direct inspection or HUMINT with physical access.

PART IV: STRATEGIC ASSESSMENT — PREWAR, WARTIME, AND POST-STRIKE

4.1 Pre-War Status (Through May 2025)

Prior to the June 2025 strikes, Iran had achieved what nonproliferation analysts termed 'nuclear latency' — possessing all components needed for rapid nuclear weapons production without having made the political decision to build one. The IAEA's June 12, 2025, declaration that Iran was in breach of its nuclear non-proliferation obligations for the first time since 2005 directly preceded the Israeli air campaign. Iran had roughly 408 kg of 60% enriched uranium, 18,000+ centrifuges installed at Natanz, 10 cascades of advanced IR-6 centrifuges at Fordow, and a breakout time estimated by the DIA at 'probably less than one week' for one bomb's worth of HEU.

4.2 Wartime and Immediate Post-Strike Status (June–November 2025)

The U.S.-Israeli strikes substantially degraded Iran's declared enrichment infrastructure. The above-ground Natanz PFEP and primary power infrastructure were destroyed. Fordow's enrichment halls were penetrated by 12 GBU-57 MOP bombs; IAEA confirmed 'almost all sensitive equipment' destroyed. Isfahan's uranium conversion line was heavily damaged. Iran fired approximately 530 ballistic missiles and 1,000+ drones at Israel and 14 missiles at U.S. assets at Al Udeid, Qatar, before a ceasefire was brokered.

Critically, the strikes did not eliminate the fissile material stockpile. IAEA Director General Grossi stated that if Iran enriched its existing stockpile to 90%, 'it would only take them a few weeks to complete the process.' The June 2025 strikes did not fully resolve the core fissile-material accumulation and verification problem.

4.3 Current Assessment (March 2026)

OBSERVED: Commercial satellite imagery indicates continued construction at Pickaxe Mountain, reconstruction activity at Taleghan-2 (Parchin), and repair work at Natanz and Isfahan. ASSESSMENT: Available reporting is consistent with a covert reconstitution posture, though the scope, location, and operational status of any rebuilt enrichment capacity is not independently verifiable. KEY UNKNOWN: Whether any centrifuge cascade is currently operational and whether the 440 kg stockpile is accessible to that cascade. IMPLICATION: The breakout timeline compresses dramatically if an operational cascade exists and the stockpile is accessible; it extends significantly if neither condition holds. Commercial satellite imagery indicates that construction and hardening activity at Pickaxe Mountain has continued and accelerated since the June 2025 strikes. Taleghan-2 at Parchin is being rebuilt. Available evidence indicates the core scientific and engineering knowledge base has not been eliminated by strikes on physical infrastructure. Iran retains a substantial population of nuclear scientists and engineers; the precise number active on weapons-relevant programs is unknown. Iran International reporting (January 2026, anonymous sources) indicates the IRGC Aerospace Force may be developing biological and chemical warhead integration for ballistic missiles as a parallel deterrence track. This report has not been independently corroborated by government or IAEA sources. The physical location of the 440 kg stockpile has not been verified by any international inspection body since the June 2025 strikes removed IAEA access.

One analytic interpretation of observed post-strike activity — consistent with Iran's historical dispersal doctrine — is that Iran may be avoiding overt reconstitution of declared facilities while seeking to establish deeper, harder-to-detect covert capability. This inference is supported by circumstantial indicators but is not confirmed. Pickaxe Mountain's assessed depth, based on satellite imagery and open-source construction timeline analysis, is consistent with exceeding the demonstrated GBU-57 penetration capability — though precise depth and interior configuration remain unknown quantities.

4.4 Final Assessment: Key Unknowns and Their Strategic Significance

CRITICAL UNKNOWN #1	*Location and integrity of 440 kg of 60% enriched uranium. This is the single most consequential unresolved intelligence question in current nonproliferation. No remote sensing method can resolve it without physical access. Every strategic option — military, diplomatic, or verification-based — is impaired without this answer.*

CRITICAL UNKNOWN #2	*Operational status of Pickaxe Mountain. If it contains active centrifuge cascades fed by diverted 60% material, Iran could be nearer the breakout threshold under worst-case assumptions than current monitoring can confirm or exclude.*

CRITICAL UNKNOWN #3	*Whether the Kavir Plan represents a real, active weaponization program in Semnan Province. If confirmed, it would indicate that Iran made the decision to actively develop nuclear weapons — a fundamental change from ODNI assessments through 2025 that Iran had not resumed the weapons program formally.*

BIOLOGICAL PROGRAM NOTE	*The January 2026 Iran International report of possible BW/CW warhead integration (Tier 3 — anonymous sources, single outlet), taken alongside the 2025 ODNI assessment upgrade to 'very likely,' may represent a parallel threat development track. The ODNI elevation rests on Tier 1 assessments but does not specify a warhead-integration dimension. Iran's framing of BW as a 'complementary deterrent factor' has been reported but not independently confirmed. This combination of indicators warrants continued monitoring; it does not constitute confirmation of an active weaponization program.*

SOURCES AND REFERENCE BASIS

All information in this assessment is drawn exclusively from open-source, publicly available materials. No classified information is contained herein.

- [TIER 1] International Atomic Energy Agency (IAEA) — Iran Safeguards Reports, 2003–2025; NPT Safeguards Agreement reports; Director General statements
- Congressional Research Service — 'Iran and Nuclear Weapons Production' (CRS IF12106, updated 2025); 'Status of Iran's Nuclear Program' reports
- [TIER 2] Institute for Science and International Security (ISIS) — Albright et al., 'Comprehensive Updated Assessment of Iranian Nuclear Sites Five Months After the 12-Day War' (Nov. 2025); 'Analysis of IAEA Iran Verification Report' (May 2025); 'Going for the Bomb: Pathways and Timelines'
- [TIER 2] Center for Strategic and International Studies (CSIS) — 'CSIS Satellite Imagery Analysis Reveals Possible Signs of Renewed Nuclear Activity in Iran' (Nov. 2025); Nuclear Network assessments
- Arms Control Association — 'Iran's Nuclear Program After the Strikes' (July 2025); 'Status of Iran's Nuclear Program' fact sheets
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